# A Visual Introduction to Parabolas

The Anatomy of a Curve

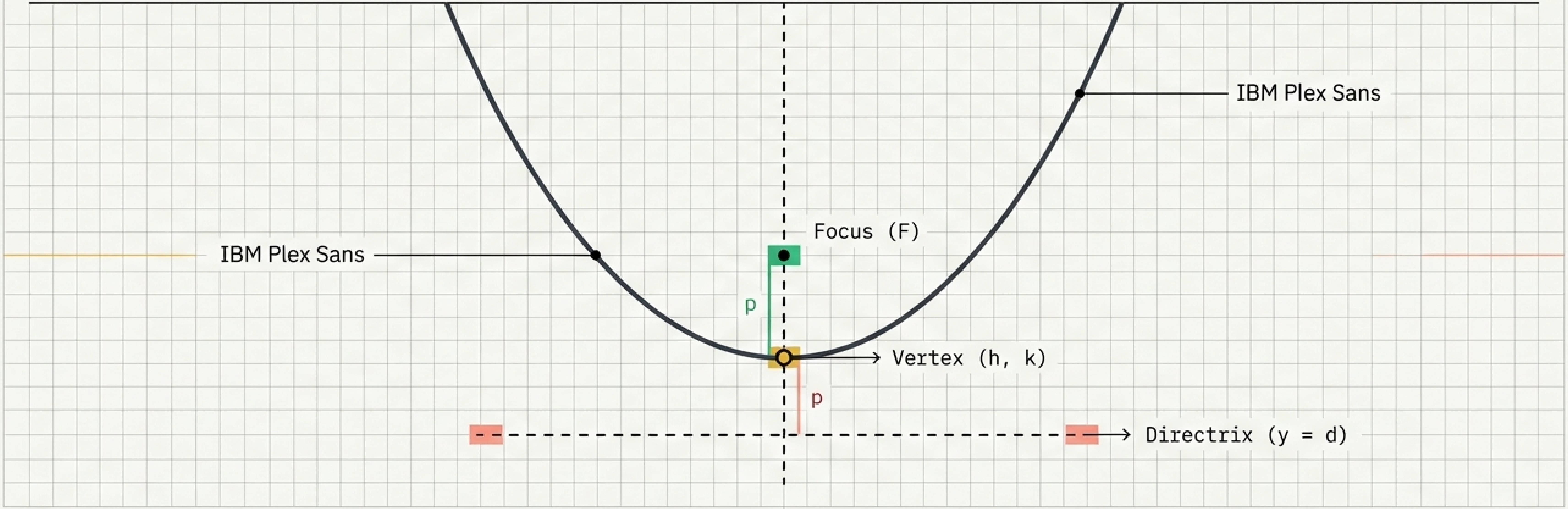
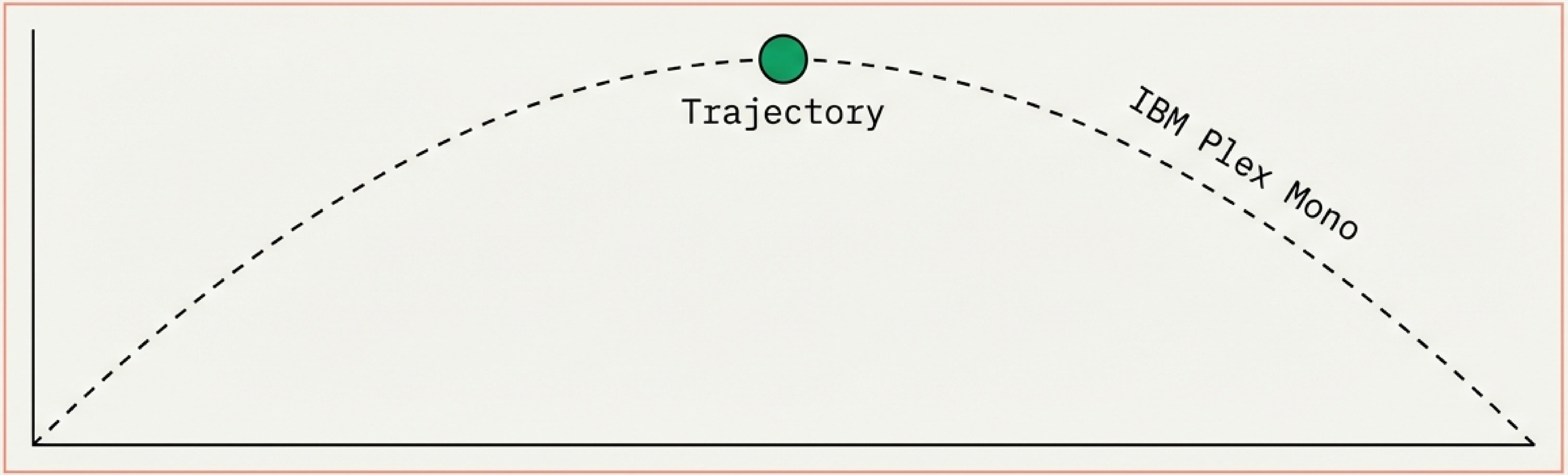



Fig 1.0: Basic Geometry

# Para (Alongside) + Bola (Thrown)

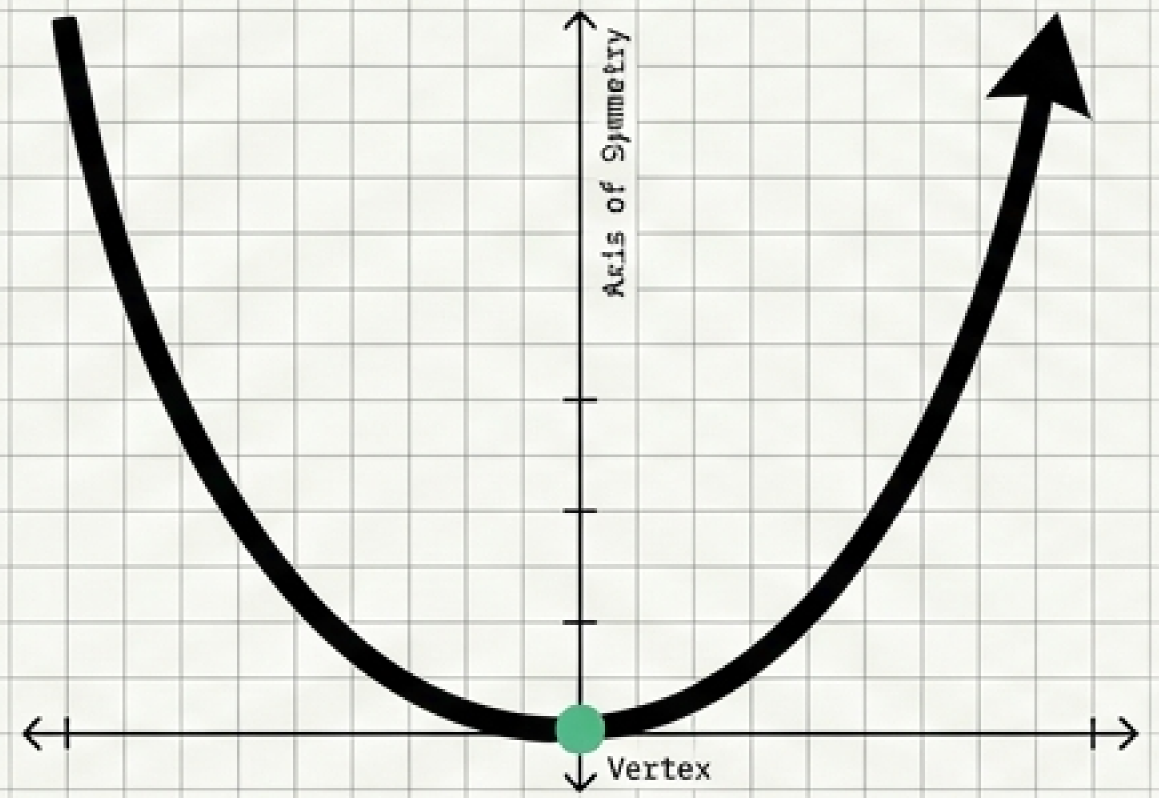
The name comes from Greek roots related to ballistics. It describes the trajectory of an object moving through the air.




**Connection:** The term shares roots with ‘Parable’ and ‘Ballistics’.

# First Impression: Orientation

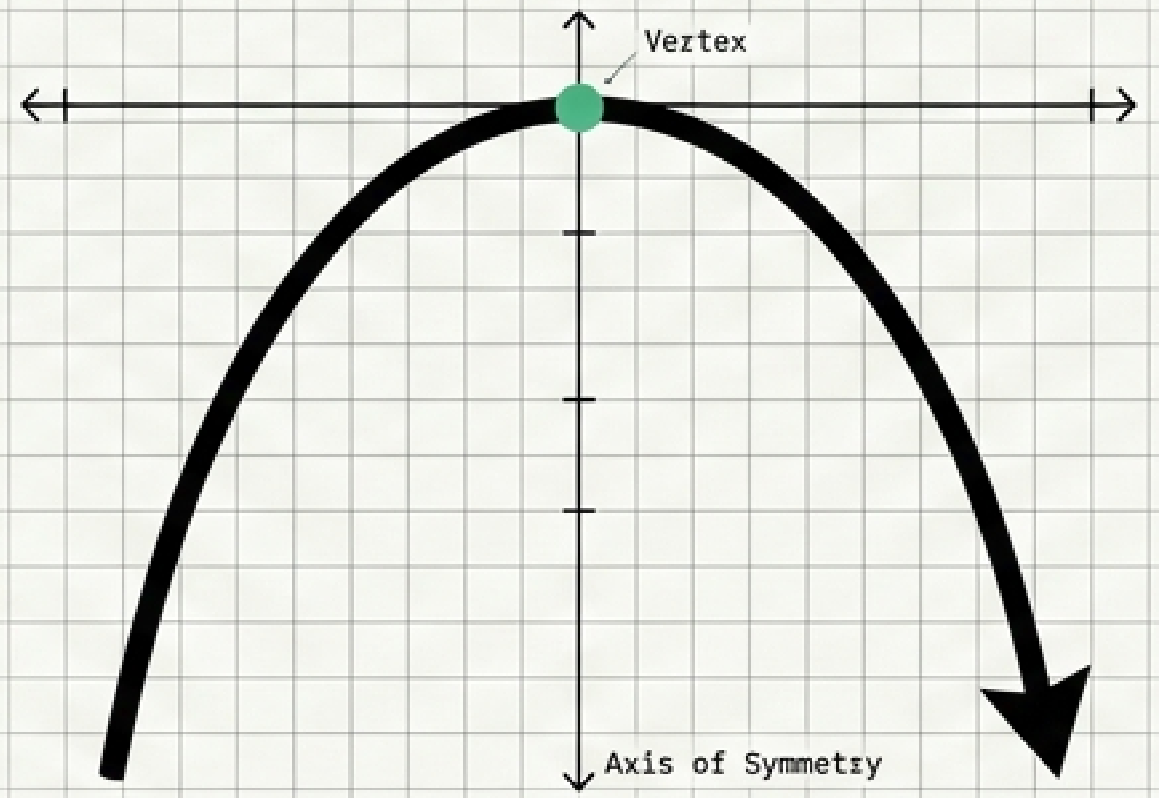
## Opens Upwards




The curve opens toward the top of the graph paper.

FIGURE 02

## Opens Downwards




The curve opens toward the bottom of the graph paper.

FIGURE 03

# The Anchor Point: The Vertex

The vertex is the distinct turning point of the parabola.

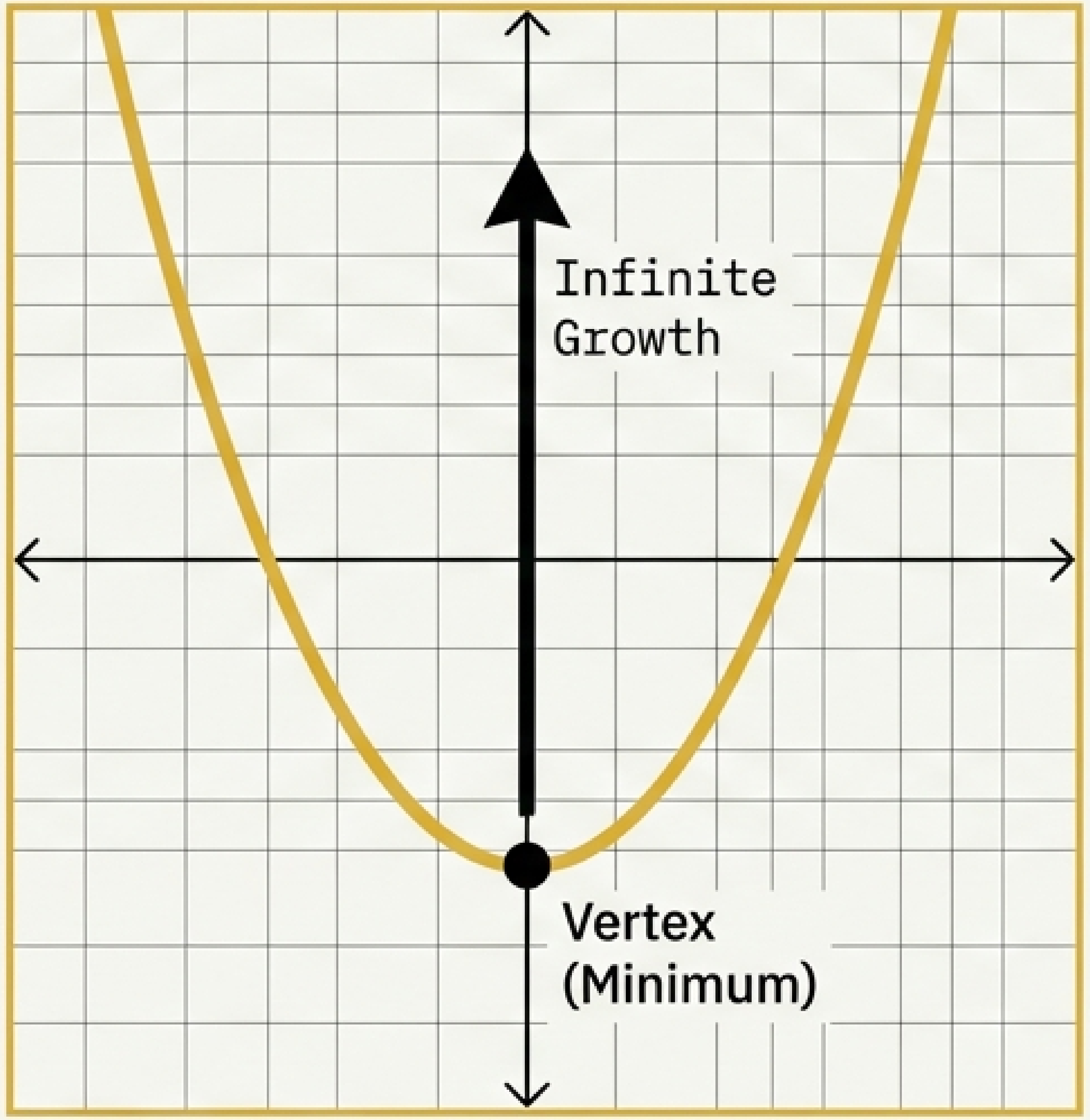



FIGURE 04

If Open Up
→ Vertex is Valley
(Minimum)

If Open Down
→ Vertex is Peak
(Maximum)

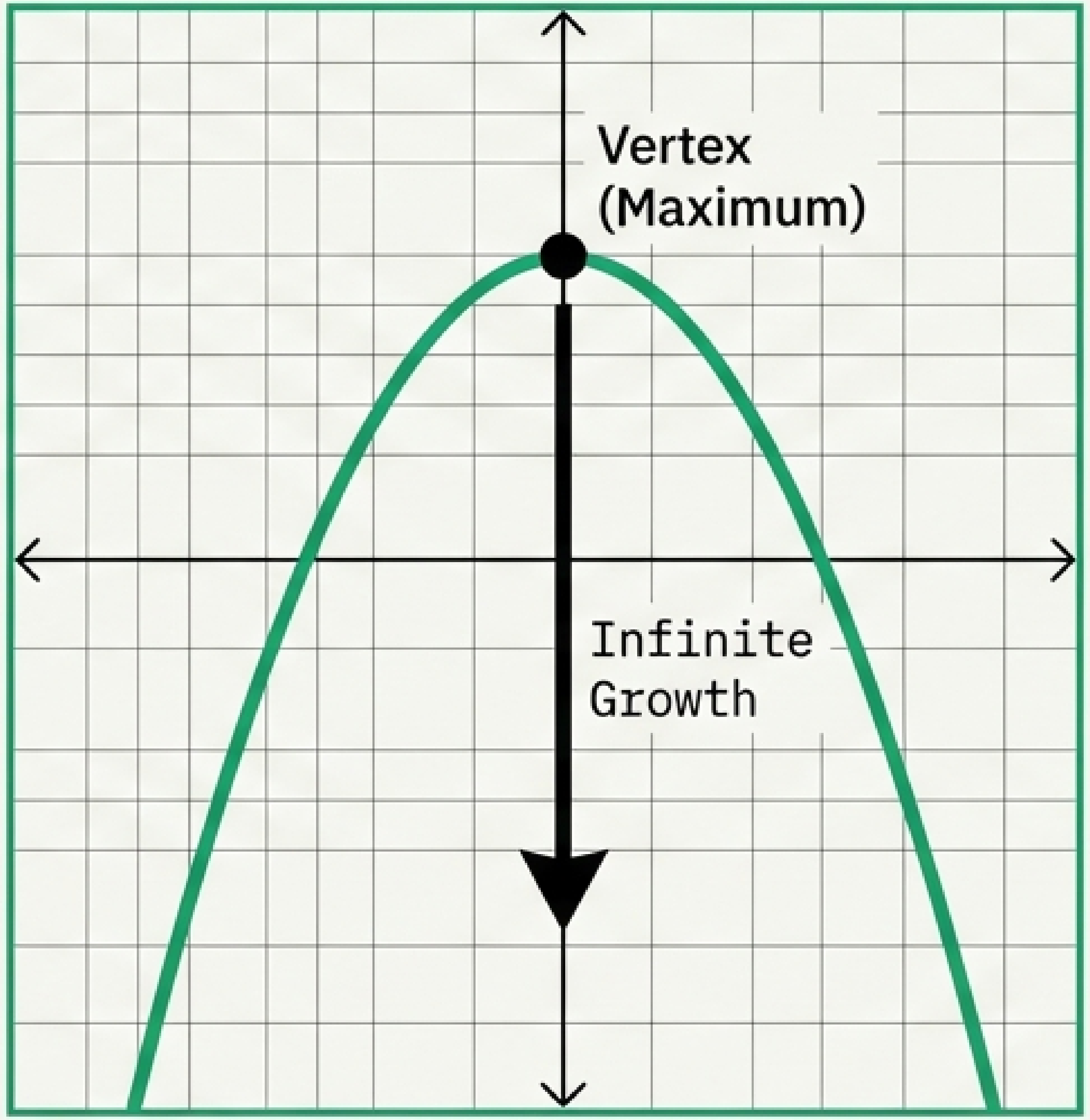



FIGURE 05

# The Mirror: Axis of Symmetry

Every parabola has a line over which you can fold the graph so it meets itself perfectly.
**This line must pass directly through the Vertex.**

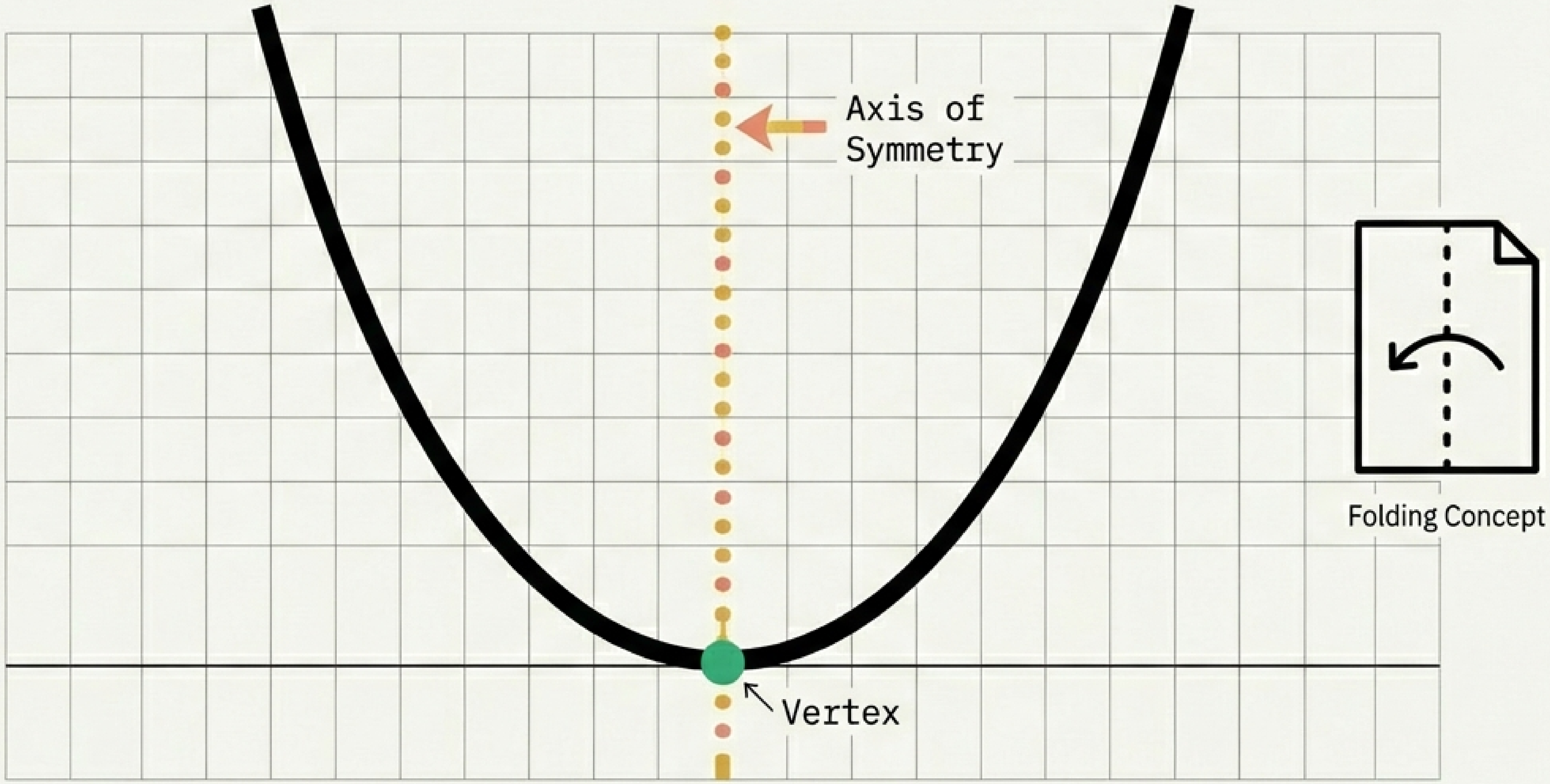
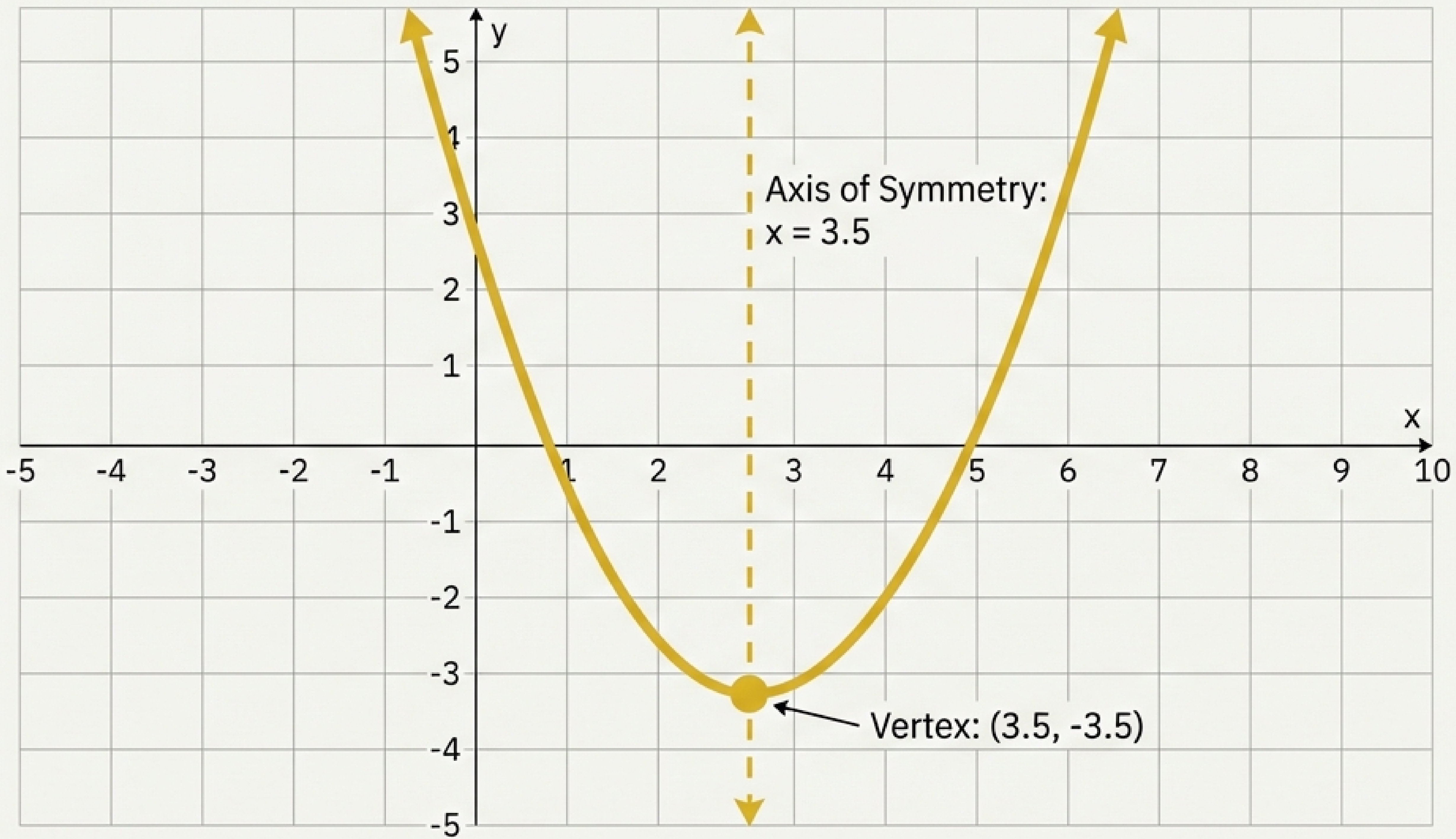



FIGURE 06

# Case Study A: The Standard Curve




**DATA SPECIFICATIONS**
in IBM Plex Mono

**Color:** Yellow

**Orientation:**
Opens Upwards

**Vertex:**
(3.5, -3.5)

**Axis of Symmetry:**
x = 3.5

**Note:** The Axis line runs directly through the x-coordinate of the Vertex.

# Connecting to the Grid: Y-Intercepts

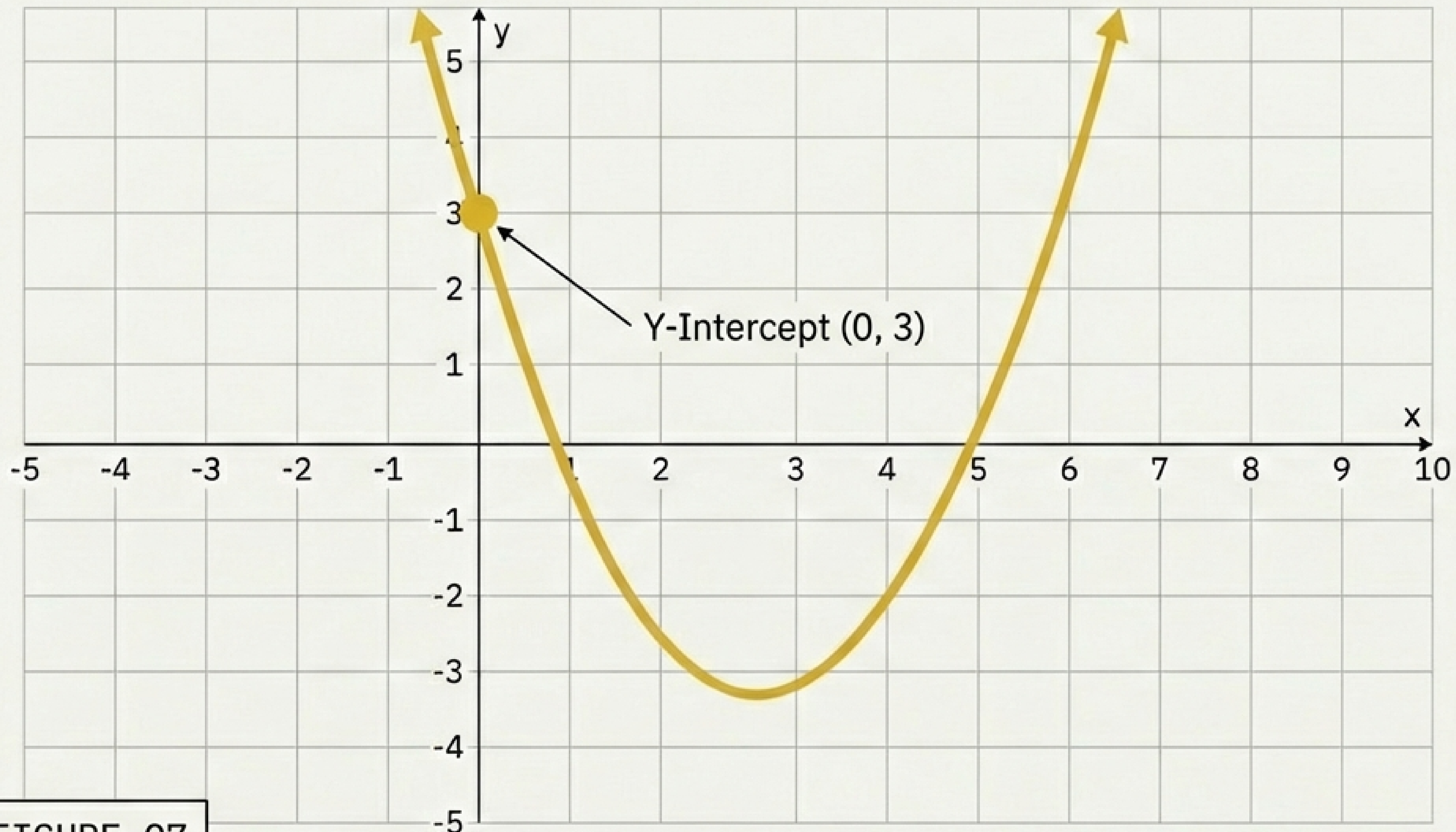



FIGURE 07

**Definition:**
The point where the curve intersects the y-axis.

**Insight:**
Every parabola will eventually hit the y-axis, even if it happens off-screen.

# The Double Hit: X-Intercepts

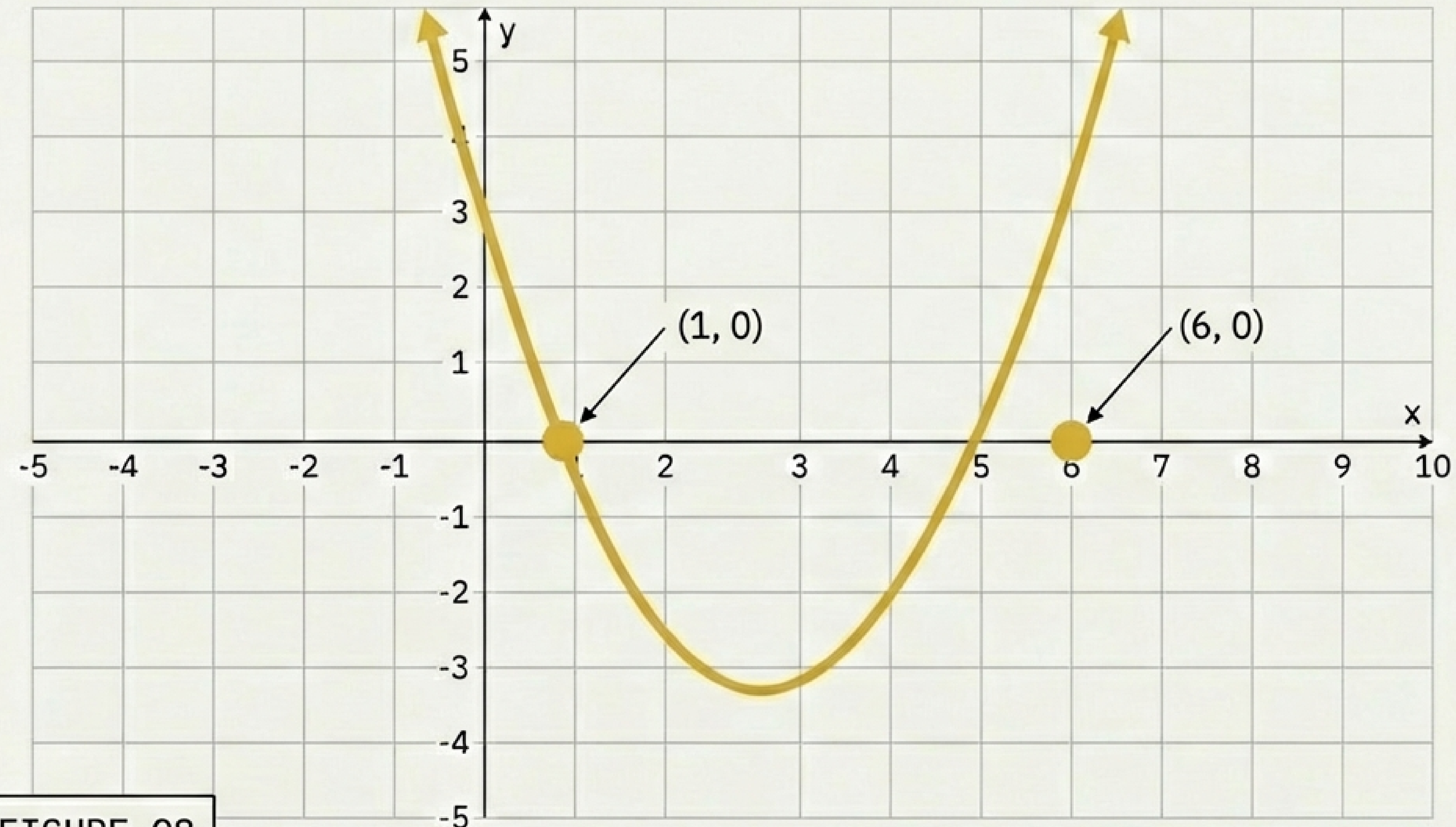
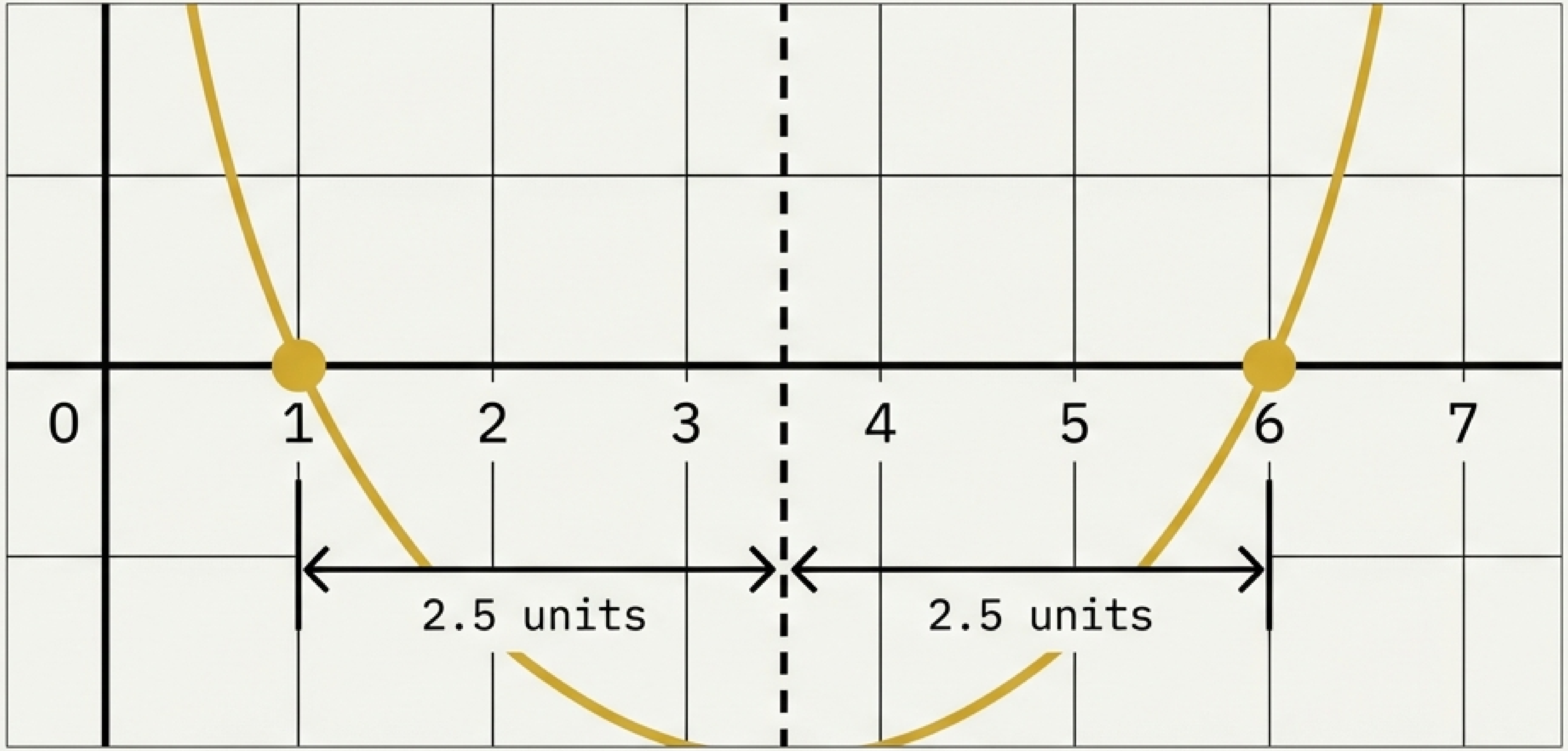



FIGURE 08

**Differentiation:**
Unlike straight lines which intersect only once, a parabola can curve back around to hit the x-axis twice.

# The Logic of Symmetry




Core Rule: X-intercepts are equidistant from the Axis of Symmetry.

Proof: The axis (3.5) is exactly in the middle of 1 and 6.

FIGURE 09

# Case Study B: The Floating Parabola

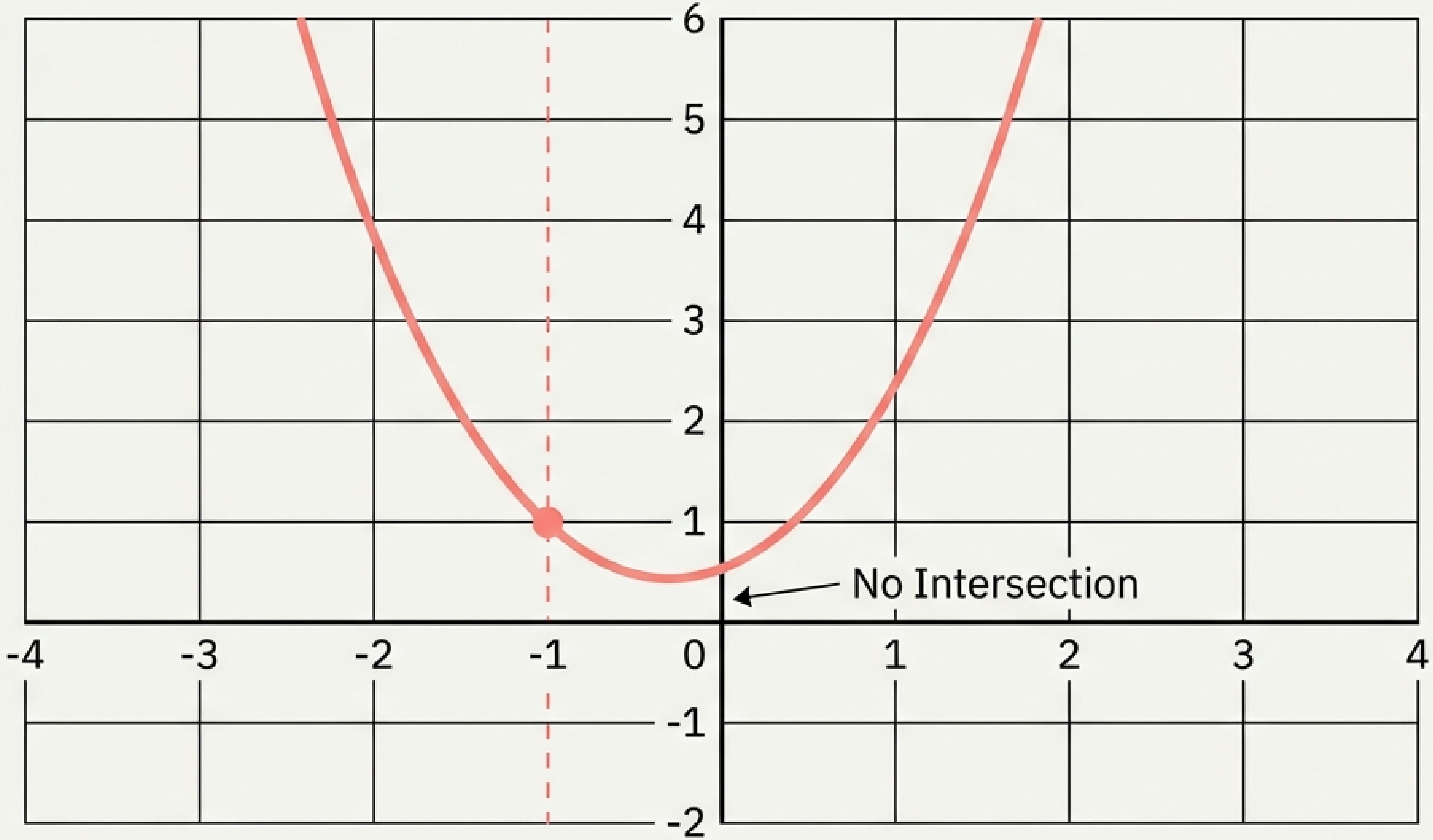



FIGURE 10

Axis of Symmetry: x = -1

Key Insight: Not all parabolas have x-intercepts.

Since the minimum point is above the axis and it opens up, it never touches the x-axis.

# Case Study C: The Downward Turn

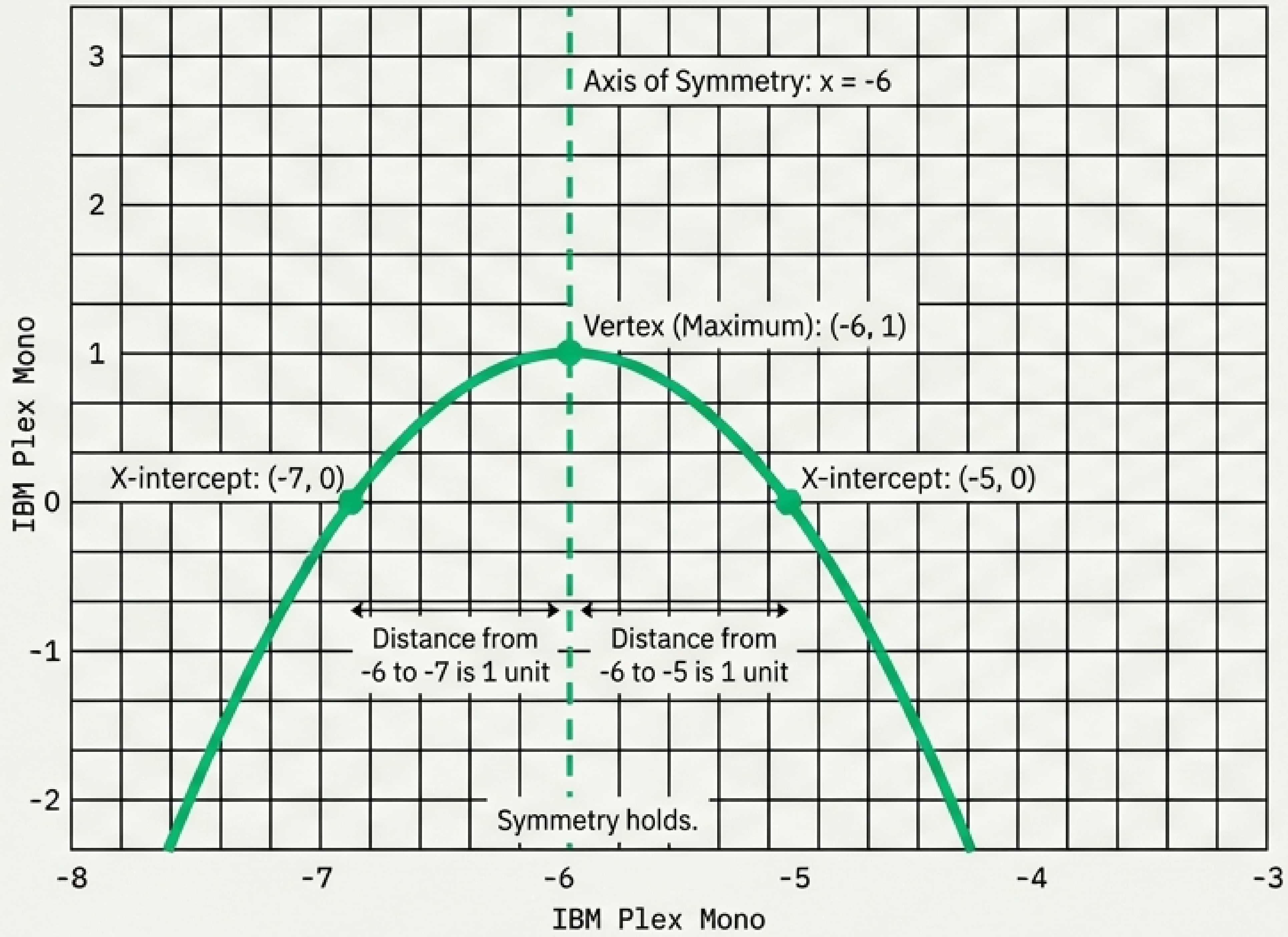
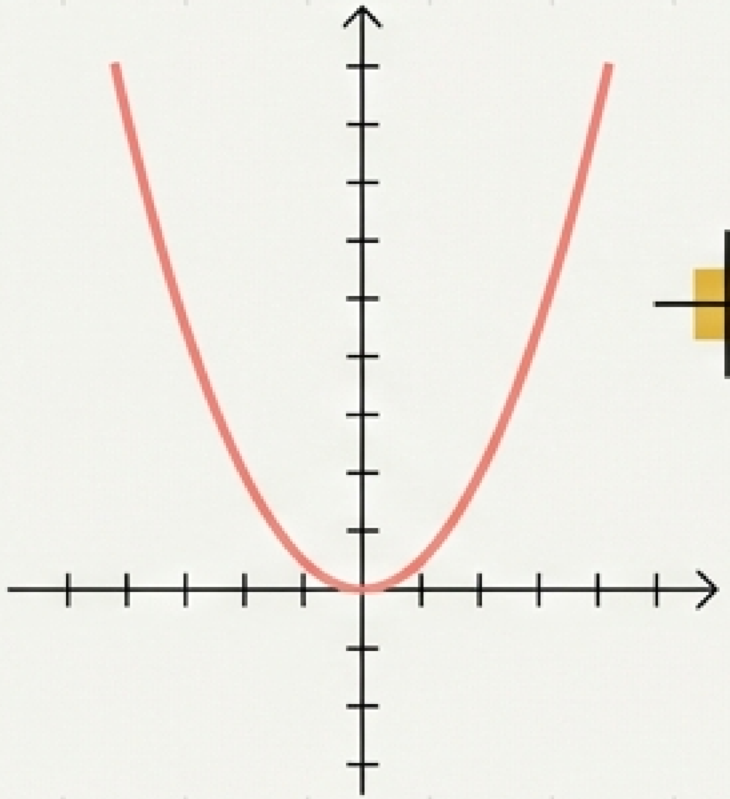



FIGURE 11

The downward curve is characterized by a negative leading coefficient.
The vertex at (-6, 1) represents the maximum value.

Both x-intercepts are exactly 1 unit away from the axis of symmetry, illustrating perfect symmetry.

# Looking Ahead: The Algebraic Connection



# Quadratics

$$y = x^2$$

Simple Equation

$$y = 2x^2 - 5x + 7$$

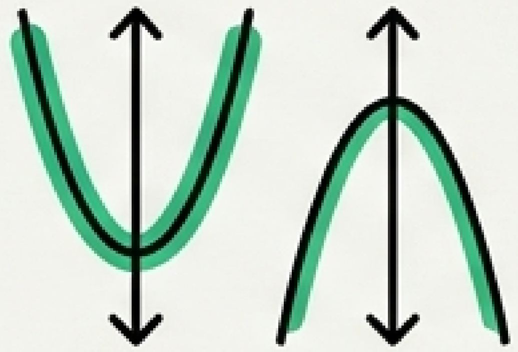
Complex Equation

FIGURE 12

The pictures we just analyzed are visual representations of equations involving second-degree terms (squared variables).

# Parabola Checklist

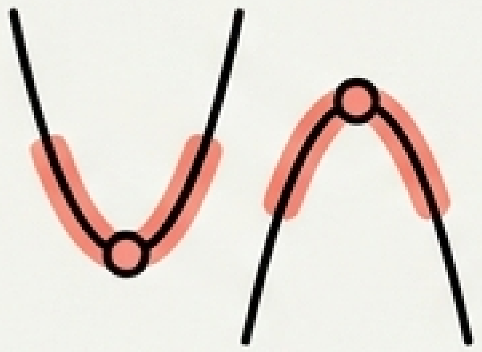
## Orientation



Opens Up (Valley) or Down (Peak)?

IBM Plex Mono →
Leading Coeff: + or -

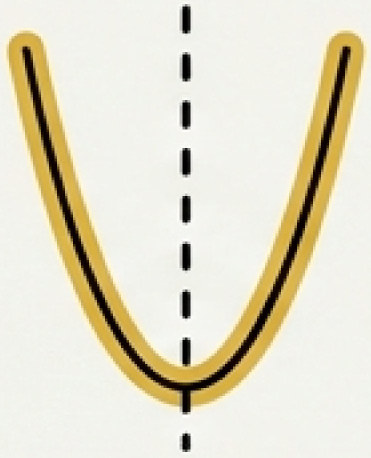
## Vertex



The Max or Min turning point.

IBM Plex Mono
(h, k) coords

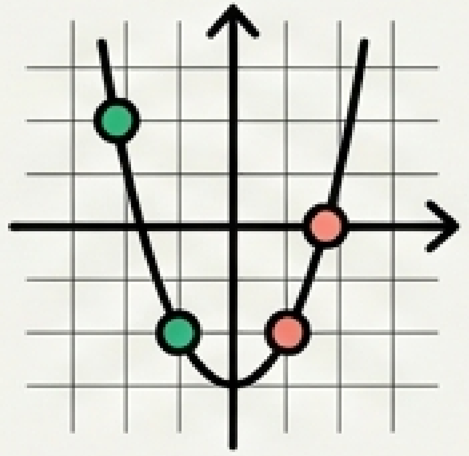
## Axis of Symmetry



The vertical mirror line through the vertex.

IBM Plex Mono
Equation: $x = h$

## Intercepts

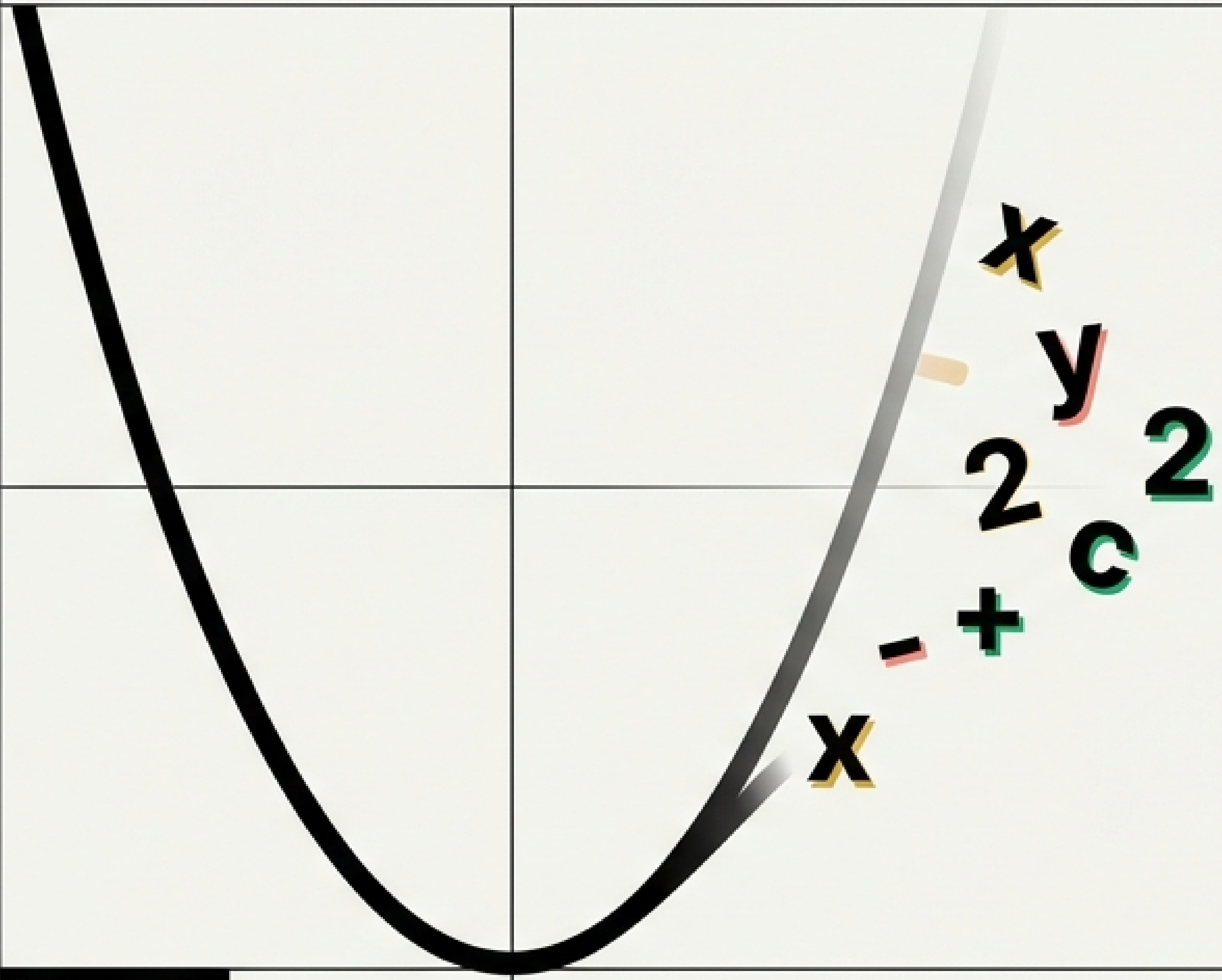


Where it hits the grid (Y is guaranteed; X is optional).

IBM Plex Mono
$(0, c)$, $(x_1, 0)$, $(x_2, 0)$

FIGURE 13

# From Vision to Calculation



$$y = ax^2 + bx + c$$

FIGURE 14

We have learned to read the shape of the curve. The next step is using algebra to calculate these points with precision.